Course name	**Network Architectures and Protocols / Computer networks**
The topic of the document	**Laboratory exercise book for university students**
Laboratory exercise book no.	**2.**
The title and the topic of the laboratory exercise	**Transmission media, network node types**

1. Preparation and implementation requirements:

- The student is obliged to read in advance the content of the present laboratory exercise book, to do the preparations for the laboratory and to observe the rules defined for the laboratory.
- The students should read the description of the special terms, mechanisms listed in the „Theoretical background information" chapter to accomplish the laboratory tasks successfully.
- The appearance happens at the beginning of the laboratory, delay can be allowed only in justified cases.
- The use of students' own mobile phone and PC is forbidden in the laboratory, except the case when the laboratory leader permits their use for the students because of the type of the laboratory exercise to be done.
- Browsers can be used exclusively for the access to information which is necessary for the accomplishment of the laboratory exercise, in the case of the laboratory leader's permission or instruction.
- By the end of the class the laboratory leader evaluates each participant's activity based on the composed questions which are included in this present material and further possible questions.
- The laboratory will be successful, if the student provides a correct answer to each question on the basis of the laboratory exercises which are carried out on the premises.
- The replacement of the laboratory is possible at dates of time determined by the laboratory leader.

2. Theoretical background information:

The ppt slides of the lecture are available at the following link:
https://arato.inf.unideb.hu/gal.zoltan/nap_ZG_2018_05_09.pdf

During the laboratory the following special terms will be used in practice:
- **Transmission medium**, **channel**: pdf slides of the lecture, slide no. 15., 58-69
- **Network Interface Card (NIC) and its modes**: Network Interface Controller – Wikipedia (wikipedia.org)
- **Physical address of the NIC** : MAC address – Wikipedia (wikipedia.org)

- **NIC modes:** Promiscuous mode – Wikipedia (wikipedia.org)
- **Cable types:** An Educator's Guide to School Networks - Ch. 4.: Cabling
- **Relationship of power ratio and decibel:** https://en.wikipedia.org/wiki/Decibel
- **Host and virtual machine**: Virtual machine – Wikipedia (wikipedia.org)
- **Network cable connector types and specifications:** https://www.computernetworkingnotes.com/networking-tutorials/network-cable-connectors-types-and-specifications.html
- **Network cable types and specifications:** https://www.computernetworkingnotes.com/networking-tutorials/network-cable-types-and-specifications.html

3. The following tasks have to be performed on the site:

3.1.	*Identification of galvanic or optical communication media and connector types*
	Identify the elements of the coaxial cable used for IEEE 802.3 technology. Perform cable tester measurements on it. Identify the elements of the twisted-pair cable. Perform cable tester measurements on it. Identify the elements of the optical fiber cable.

3.2.	*Performance of mental calculation between power ratio (x) and decibel (dB(x)).*
	On the basis of your manual calculation determine the value of power ratios x = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 in dBs.

3.3.	*Introduction to Oracle VM VirtualBox environment*
	Download of Knoppix software, creation and configuration of a virtual machine (VM). Creating a clone of a VM. Power off VM options. Network interfaces. Directory structure on the host machine.

4. The following questions have to be answered by the end of the laboratory:

4.1.	A) Characteristics of coaxial cable: - diameter [mm]: - list of connector types - length of connector (coupler) to NIC [mm]:
	B) Characteristics of twisted-pair cable: - diameter [mm]: - list of connector components - length of plug to NIC [mm]:
	C) Characteristics of optical fiber cable: - diameter of outer plastic jacket [mm]: - list of cable types according to length in increasing order:

	- diameter of plug (ceramic, metal) submodule [mm]:

4.2.	A) On the basis of your mental calculation how many decibel values do correspond to the following power ratios (x)? x = 12 in dBs: x = 14 in dBs: x = 16 in dBs:
	B) On the basis of your mental calculation how many values of power ratio do correspond to the following decibels? In the case of 13 dBs x = In the case of -68 dBs x = In the case of -70 dBs x =

4.3.	A) The properties of the Oracle VM VirtualBox - version number - number of possible VM types - possible types of VM network interfaces - possible number of VM network interfaces
	B) The file structure properties of the Oracle VM VirtualBox - relationship between the file structure of the VM and that of the host (paths)